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Polycystic Ovarian Syndrome

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Overview

Background

Patients with polycystic ovarian syndrome (PCOS) have abnormalities in the metabolism of androgens and estrogen and in the control of androgen production.[1] The major features of PCOS include menstrual dysfunction, anovulation, and signs of hyperandrogenism.

Although the exact etiopathophysiology of this condition is unclear, PCOS can result from abnormal function of the hypothalamic-pituitary-ovarian (HPO) axis. A key characteristic of PCOS is inappropriate gonadotropin secretion, which is more likely a result of, rather than a cause of, ovarian dysfunction. In addition, one of the most consistent biochemical features of PCOS is a raised plasma testosterone level.

Stein and Leventhal were the first to recognize an association between the presence of polycystic ovaries and signs of hirsutism and amenorrhea (eg, oligomenorrhea, obesity).[2] After women diagnosed with Stein-Leventhal syndrome underwent successful wedge resection of the ovaries, their menstrual cycles became regular, and they were able to conceive.[3] As a consequence, a primary ovarian defect was thought to be the main culprit, and the disorder came to be known as polycystic ovarian disease.

Further biochemical, clinical, and endocrinologic studies revealed an array of underlying abnormalities. As a result, the condition is now referred to as PCOS, although it may occur in women without ovarian cysts and although ovarian morphology is no longer an essential requirement for diagnosis.

A diagnosis of polycystic ovarian morphology (as opposed to PCOS) is defined in adults by the presence of 20 or more follicles per ovary and/or an ovarian volume of at least 10 mL in either ovary, using high-frequency (transvaginal) ultrasonography. If older technology or transabdominal imaging is used, the threshold for follicle count should be increased or the diagnosis should rely on ovarian volume.[1] See the images below.

Diagnostic criteria

A 1990 expert conference sponsored by the National Institute of Child Health and Human Disease (NICHD) of the United States National Institutes of Health (NIH) proposed the following criteria for the diagnosis of PCOS[4]:

- Oligo-ovulation or anovulation manifested by oligomenorrhea or amenorrhea
- Hyperandrogenism (clinical evidence of androgen excess) or hyperandrogenemia (biochemical evidence of androgen excess)
- Exclusion of other disorders that can result in menstrual irregularity and hyperandrogenism

In 2003, the European Society for Human Reproduction and Embryology (ESHRE) and the American Society for Reproductive Medicine (ASRM) recommended that at least two of the following three features (the Rotterdam criteria) are required for PCOS to be diagnosed[5]:

- Oligo-ovulation or anovulation manifested as oligomenorrhea or amenorrhea
- Hyperandrogenism (clinical evidence of androgen excess) or hyperandrogenemia (biochemical evidence of androgen excess)
- Polycystic ovaries (as defined on ultrasonography)

A research analysis by Copp et al pointed out that since the expanded criteria for PCOS diagnosis from the Rotterdam consensus, the estimated number of diagnoses in women of reproduction age increased from 4-6.6% to 21%.[6]

The Androgen Excess and PCOS Society (AE-PCOS) published a position statement emphasizing that, in the society's opinion, PCOS should be considered a disorder of androgen excess, as defined by the following[7,8]:

- Clinical/biochemical evidence of hyperandrogenism
- Evidence of ovarian dysfunction (oligo-ovulation and/or polycystic ovaries)
- Exclusion of related disorders

The Society of Obstetricians and Gynaecologists of Canada (SOGC) indicated that a diagnosis of PCOS can be made in the presence of at least two of the following three criteria, when congenital adrenal hyperplasia, androgen-secreting tumors, or Cushing syndrome have been excluded[9]:

- Oligo-ovulation or anovulation
- Clinical/biochemical evidence of hyperandrogenism
- Polycystic ovaries on ultrasonography

In 2023, the International PCOS Network released evidence-based guideline criteria, which require the presence of two of the following to diagnose PCOS[1]:

- Clinical and/or biochemical evidence of hyperandrogenism
- Ovulatory dysfunction
- Polycystic ovaries as indicated by ultrasound examination or anti-Müllerian hormone (AMH) level in adults

Neither ultrasonography nor measurement of serum AMH is recommended in adolescents because of poor specificity in this age group.

PCOS can be diagnosed in patients who have both irregular menstrual cycles and hyperandrogenism. In such patients, neither an ovarian ultrasound scan nor an AMH level is required for the diagnosis.

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Etiology

High serum concentrations of androgenic hormones, such as testosterone, androstenedione, and dehydroepiandrosterone sulfate (DHEA-S), may be encountered in patients with PCOS. However, individual variation is considerable, and a particular patient might have normal androgen levels.

PCOS is also associated with peripheral insulin resistance and hyperinsulinemia, and obesity amplifies the degree of both abnormalities. Insulin resistance in PCOS can be secondary to a postbinding defect in insulin receptor signaling pathways, and elevated insulin levels may have gonadotropin-augmenting effects on ovarian function. Hyperinsulinemia may also result in suppression of hepatic generation of sex hormone–binding globulin (SHBG), which in turn may increase androgenicity.[10]

In addition, insulin resistance in PCOS has been associated with adiponectin, a hormone secreted by adipocytes that regulates lipid metabolism and glucose levels. Lean and obese women with PCOS have lower adiponectin levels than do women without PCOS.[11]

A proposed mechanism for anovulation and elevated androgen levels suggests that, under the increased stimulatory effect of luteinizing hormone (LH) secreted by the anterior pituitary, stimulation of the ovarian theca cells is increased. These cells, in turn, increase the production of androgens (eg, testosterone, androstenedione). Because of a decreased level of follicle-stimulating hormone (FSH) relative to LH, the ovarian granulosa cells cannot aromatize the androgens to estrogens, which leads to decreased estrogen levels and consequent anovulation. Growth hormone (GH) and insulin-like growth factor–1 (IGF-1) may also augment the effect on ovarian function.[12]

Hyperinsulinemia is also responsible for dyslipidemia and for elevated levels of plasminogen activator inhibitor-1 (PAI-1) in patients with PCOS. Elevated PAI-1 levels are a risk factor for intravascular thrombosis.

Polycystic ovaries are enlarged bilaterally and have a smooth, thickened capsule that is avascular. On cut sections, subcapsular follicles in various stages of atresia are seen in the peripheral part of the ovary. The most striking ovarian feature of PCOS is hyperplasia of the theca stromal cells surrounding arrested follicles. On microscopic examination, luteinized theca cells are seen.

Some evidence suggests that patients have a functional abnormality of cytochrome P450c17, the 17-hydroxylase, which is the rate-limiting enzyme in androgen biosynthesis.[11]

PCOS is a genetically heterogeneous syndrome in which the genetic contributions remain incompletely described. PCOS is an inherently difficult condition to study genetically because of its heterogeneity, difficulty with retrospective diagnosis in

postmenopausal women, associated subfertility, incompletely understood etiology, and gene effect size. Many published genetics studies in PCOS have been underpowered, and the results of published candidate gene studies have been disappointing.

Studies of family members with PCOS indicate that an autosomal dominant mode of inheritance occurs for many families with this disease. The fathers of women with PCOS can be abnormally hairy; female siblings may have hirsutism and oligomenorrhea; and mothers may have oligomenorrhea.[13] Research has suggested that in a large cohort of women with PCOS, a family history of type 2 diabetes in a first-degree family member is associated with an increased risk of metabolic abnormality, impaired glucose tolerance, and type II diabetes.[13] In addition, a Dutch twin-family study showed a PCOS heritability of 0.71 in monozygotic twin sisters, versus 0.38 in dizygotic twins and other sisters.[14]

An important link between PCOS and obesity was corroborated genetically by data from a case-control study in the United Kingdom that involved 463 patients with PCOS and more than 1300 female controls.[15] The investigators demonstrated that a variant within the FTO gene (rs9939609, which has been shown to predispose to common obesity) was significantly associated with susceptibility to the development of PCOS.

Wickenheisser et al reported that CYP17 promoter activity was 4-fold greater in cells of patients with PCOS. This research suggests that the pathogenesis of PCOS may be in part related to the gene regulation of CYP17.[16] However, in a study that assessed candidate genes for PCOS using microsatellite markers to look for association in 4 genes— CYP19, CYP17, FST, and INSR —only 1 marker near the INSR gene was found to be significantly associated with PCOS.[17] The authors concluded that a susceptibility locus for PCOS (designated PCOS1) exists in 19p13.3 in the INSR region, but it cannot be concluded that the INSR gene itself is responsible.[17]

Subsequent studies have found additional associations, such as those of 15 regions in 11 genes previously described to influence insulin resistance, obesity, or type 2 diabetes.[18] Individuals with PCOS were found more likely to be homozygous for a variant upstream of the PON1 gene and homozygous for an allele of interest in IGF2. Interestingly, the PON1 gene variant resulted in decreased gene expression, which could increase oxidative stress. The exact result of the IGF2 variant is unclear, but IGF2 stimulates androgen secretion in the ovaries and adrenal glands.[18]

In study by Goodarzi et al, the leucine allele was found to be associated with protection against PCOS, as compared to the valine allele at position 89 in SRD5A2.[19] The leucine allele is associated with a lower enzyme activity.[19] When the results of this study are combined with those of an observational study by Vassiliadi et al, based on urinary steroid profiles in women with PCOS, further support can be found for an important role for 5-alpha reductase in the pathogenesis of this syndrome.[20]

In a genome-wide association study for PCOS in a Han Chinese population, 3 strong regions of association were identified, at 2p16.3, 2p21, and 9q33.3.[21] The polymorphism most strongly associated with PCOS at the 2p16 locus was near several genes involved in proper formation of the testis, as well as a gene that encodes a receptor for luteinizing hormone (LH) and human chorionic gonadotropin (HCG). This polymorphism was also located 211kb upstream from the FSHR gene, which encodes the follicle-stimulating hormone (FSH) receptor.[21]

The polymorphisms most strongly associated with PCOS at the 2q21 locus encode a number of genes, including the THADA gene, which has previously been associated with type 2 diabetes. In addition, 6 significant polymorphisms were identified as being associated with PCOS at the 9q33.3 locus near the DENND1A gene, which interacts with the ERAP1 gene. Elevation in serum ERAP1 has been previously associated with PCOS and obesity.[21]

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Epidemiology

In the United States, PCOS is one of the most common endocrine disorders of reproductive-age women, with a prevalence of 4-12%.[22,23] Up to 10% of women are diagnosed with PCOS during gynecologic visits.[24] In some European studies, the prevalence of PCOS has been reported to be 6.5-8%.[25,26]

A great deal of ethnic variability in hirsutism is observed. For example, Asian (East and Southeast Asia) women have less hirsutism than White women given the same serum androgen values. In a study that assessed hirsutism in southern Chinese women, investigators found a prevalence of 10.5%.[27] In hirsute women, there was a significant increase in the incidence of acne, menstrual irregularities, polycystic ovaries, and acanthosis nigricans.[27]

PCOS affects premenopausal women, and the age of onset is most often perimenarchal (before bone age reaches 16 years). However, clinical recognition of the syndrome may be delayed by failure of the patient to become concerned by irregular menses, hirsutism, or other symptoms or by the overlap of PCOS findings with normal physiologic maturation during the 2 years after menarche. In lean women with a genetic predisposition to PCOS, the syndrome may be unmasked when they subsequently gain weight.[10]

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Prognosis

Cardiovascular and cerebrovascular disease

Evidence suggests that women with PCOS are at increased risk for cardiovascular and cerebrovascular disease. Women with hyperandrogenism have elevated serum lipoprotein levels similar to those of men.[28,29]

A meta-analysis that included 1.06 million women demonstrated that PCOS is associated with a significantly higher risk of composite cardiovascular disease (odds ratio [OR], 1.68), composite ischemic heart disease (OR, 1.48), myocardial infarction (OR, 2.50), and stroke (OR, 1.71).[30] A study by Zhao et al showed that even young, nonobese women with PCOS are at increased risk for long-term cardiovascular disease events, compared with women who do not have the syndrome.[31]

In a study in Nordic countries, the risk for cardiovascular disease, pulmonary embolism or deep vein thrombosis, and major adverse cardiovascular events was 27%, 30%, and 21% higher, respectively, among women with PCOS than among women without PCOS. Moreover, the risk for cardiovascular disease was 40% higher among women with PCOS who had a body mass index (BMI) lower than 25 and no diagnosis of type 2 diabetes, compared with a control group.[32]

Mean platelet volume (MPV) is a marker associated with adverse cardiovascular events, and women with newly diagnosed PCOS appear to have significantly elevated MPV levels.[33] Although MPV is recognized as a surrogate marker of platelet activation and prothrombotic risk, its clinical utility in PCOS remains investigational.

Type 2 diabetes mellitus

Approximately 40% of patients with PCOS have insulin resistance that is independent of body weight. These women are at increased risk for type 2 diabetes mellitus and consequent cardiovascular complications. The American Association of Clinical Endocrinologists (AACE) and international consensus guidelines recommend screening for impaired glucose tolerance and type 2 diabetes mellitus at the time of initial PCOS diagnosis in all adolescents and adults, regardless of BMI.[1,34]

Endometrial hyperplasia and carcinoma

Patients with PCOS are also at an increased risk for endometrial hyperplasia and carcinoma.[35] The chronic anovulation in PCOS leads to constant endometrial stimulation with estrogen without progesterone, and this increases the risk of endometrial hyperplasia and carcinoma.

No known association with breast or ovarian cancer has been found; thus, no additional surveillance is needed.

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Presentation

History

The family history of patients with polycystic ovarian syndrome (PCOS) may include the following:

- Menstrual disorders
- Adrenal enzyme deficiencies
- Hirsutism
- Infertility
- Obesity and metabolic syndrome
- Diabetes

Menstrual abnormalities

Patients with PCOS have abnormal menstruation patterns attributed to chronic anovulation. The patient usually has a history of menstrual disturbance dating back to menarche. Some women have oligomenorrhea (ie, menstrual bleeding that occurs at intervals of 35 days to 6 months, with < 9 menstrual periods per year) or secondary amenorrhea (an absence of menstruation for 6 months). Dysfunctional uterine bleeding and infertility are the other consequences of anovulatory menstrual cycles. The menstrual irregularities in PCOS usually present around the time of menarche.

A retrospective study by Maslyanskaya et al reported that PCOS was the most common etiology seen in adolescent patients hospitalized for abnormal uterine bleeding (accounting for 33% of 125 hospital admissions).[36]

Hyperandrogenism

Hyperandrogenism clinically manifests as excess terminal body hair in a male distribution pattern. Hair is commonly seen on the upper lip, on the chin, around the nipples, and along the linea alba of the lower abdomen. Some patients have acne and/or male-pattern hair loss (androgenic alopecia).

Other signs of hyperandrogenism (eg, clitoromegaly, increased muscle mass, voice deepening) are more characteristic of an extreme form of PCOS termed hyperthecosis. These signs and symptoms could also be consistent with androgen-producing tumors, exogenous androgen administration, or virilizing congenital adrenal hyperplasia.

Premature adrenarche is a common occurrence and, in some cases, may represent a precursor to PCOS. Hirsutism and obesity may be present in premenarchal adolescent girls with PCOS.

The American College of Obstetricians and Gynecologists (ACOG) recommends screening with 17-hydroxyprogesterone levels in women with suspected PCOS who are at increased risk for nonclassical congenital adrenal hyperplasia.[37]

Infertility

A subset of women with PCOS is infertile. Most women with PCOS ovulate intermittently. Conception may take longer than in other women, or women with PCOS may have fewer children than they had planned. In addition, the rate of miscarriage is also higher in affected women.

Obesity, metabolic syndrome, and atherosclerosis

Nearly half of all women with PCOS are clinically obese. A study comparing the body mass index (BMI) in American and Italian women with PCOS showed that American women had a BMI higher than that of their Italian counterparts.[38] Women with PCOS should be assessed for their cardiovascular risk by evaluating their BMI, fasting lipid and lipoprotein levels, and risk factors for metabolic syndrome.[37]

Many patients with PCOS have characteristics of metabolic syndrome; one study showed a 43% prevalence of metabolic syndrome in women with PCOS.[22] In women, metabolic syndrome is characterized by abdominal obesity (waist circumference >35 in), dyslipidemia (triglyceride level >150 mg/dL, high-density lipoprotein cholesterol [HDL-C] level < 50 mg/dL), elevated blood pressure, a proinflammatory state characterized by an elevated C-reactive protein level, and a prothrombotic state characterized by elevated plasminogen activator inhibitor-1 (PAI-1) and fibrinogen levels.[22]

Women with PCOS have a significantly higher prevalence of subclinical atherosclerosis, as evidenced by increased coronary artery calcification and carotid intima-media thickness, than women without the syndrome.[39]

Diabetes mellitus

ACOG recommends screening for type 2 diabetes mellitus and impaired glucose tolerance in women with PCOS by obtaining a fasting glucose level and then a 2-hour glucose level after a 75-g glucose load.[37] Approximately 10% of women with PCOS have type 2 diabetes mellitus, and 30-40% of women with PCOS have impaired glucose tolerance by 40 years of age.[40,41]

Sleep apnea

Many women with PCOS have obstructive sleep apnea syndrome (OSAS), which is an independent risk factor for cardiovascular disease. Ask these patients and/or their partners about excessive daytime somnolence; individuals with obstructive sleep apnea experience apnea/hypopnea episodes during sleep.[42] For women with PCOS with suspected OSAS, there should be a low threshold for referral for sleep assessment. Patients may also be screened for OSAS in the clinic using such tools as the Epworth sleepiness score.

Depression and anxiety disorders

International evidence-based guidelines recommend screening for depression and anxiety disorders among patients with PCOS.[1] A prospective case-control study showed that women with PCOS have significantly higher levels of psychological symptoms, including depression, anxiety, and elevated perceived stress, compared to women without PCOS.[43]

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Physical Examination

Hirsutism and virilizing signs

Patients may have excessive body hair in a male distribution pattern, as well as acne. Some patients have virilizing signs, such as male-pattern balding or alopecia, increased muscle mass, deepening voice, or clitoromegaly; these findings should prompt a search for other causes of hyperandrogenism.

The modified Ferriman-Gallwey (mFG) score grades 9 body areas from 0 (no hair) to 4 (frankly virile), including the upper lip, chin, chest, upper abdomen, lower abdomen, thighs, back, arm, and buttocks. A total score of 8 or more is considered abnormal for an adult white woman; a score of 36 is the most severe.

Obesity

Approximately 50% of women with polycystic ovarian syndrome (PCOS) have abdominal obesity, characterized by a waist circumference greater than 35 inches (>88 cm).

Acanthosis nigricans

Acanthosis nigricans is a diffuse, velvety thickening and hyperpigmentation of the skin. It may be present at the nape of the neck, axillae, area beneath the breasts, intertriginous areas, and exposed areas (eg, elbows, knuckles). In patients with PCOS, acanthosis nigricans is thought to be the result of insulin resistance, although syndromic and familial variants are described. Acanthosis nigricans can also be a cutaneous marker of malignancy.

Acanthosis nigricans is staged according to the scoring system below:

- Absent (0): Not detectable on close inspection
- Present (1): Clearly present on close visual inspection, not visible to the casual observer, extent not measurable
- Mild (2): Limited to the base of the skull, usually does not extend to the lateral margins of the neck
- Moderate (3): Extends to the lateral margins of the neck but not visible anteriorly
- Severe (4): Visible anteriorly
- Severe (5): Circumferential

Blood pressure

Patients with signs and symptoms of metabolic syndrome may have elevated blood pressure, with a systolic blood pressure of 130 mm Hg or higher and a diastolic blood pressure of 85 mm Hg or higher.

Enlarged ovaries

Enlarged ovaries may not always be present. Evaluate for an ovarian mass.

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Diagnostic Considerations

All conditions that mimic polycystic ovarian syndrome (PCOS) should be ruled out before a diagnosis of PCOS is confirmed. Consider the following in the differential diagnosis of PCOS:

- Congenital adrenal hyperplasia (late-onset)
- Cushing syndrome (low potassium, striae, central obesity, high cortisol; high androgen levels in adrenal carcinoma)
- Drugs (eg, danazol, androgenic progestins)
- Exogenous anabolic steroid use
- Familial hirsutism
- Hyperprolactinemia
- Hypothyroidism
- Idiopathic hirsutism
- Masculinizing tumors of the adrenal gland or ovary (rapid onset of signs of virilization)
- Ovarian hyperthecosis
- Stromal hyperthecosis (valproic acid)

Although obesity itself is not considered part of the differential diagnosis, obesity is associated with insulin resistance or any condition that is associated with severe insulin resistance (eg, insulin receptoropathies), which may clinically manifest in the same way as PCOS. Obesity may unmask features of PCOS in women who are genetically predisposed to this syndrome.

Differential Diagnoses

- 3-Beta-Hydroxysteroid Dehydrogenase Deficiency
- Congenital Adrenal Hyperplasia
- Gigantism and Acromegaly
- Hyperprolactinemia
- Hyperthyroidism and Thyrotoxicosis
- Hypothyroidism
- Iatrogenic Cushing Syndrome
- Ovarian Cancer

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Workup

Workup

Approach Considerations

The diagnosis of polycystic ovarian syndrome (PCOS) requires the exclusion of all other disorders that can result in menstrual irregularity and hyperandrogenism, including adrenal or ovarian tumors, thyroid dysfunction, congenital adrenal hyperplasia, hyperprolactinemia, acromegaly, and Cushing syndrome. Biochemical and/or imaging studies must be done to rule out these other possible disorders and ascertain the diagnosis. A karyotype usually excludes mosaic Turner syndrome as a cause of the primary amenorrhea.

The following baseline screening tests are recommended for women with suspected PCOS[1]:

- Thyroid function tests
- Serum prolactin levels
- Free androgen index (defined as total testosterone divided by sex hormone–binding globulin [SHBG] × 100, to give a calculated free testosterone level)

Patients who are having difficulty conceiving should receive an adequate workup, along with their partners, to rule out factors that might contribute to infertility.

Samples for laboratory studies should be drawn early in the morning, with the patient in a fasting state; in women with regular menses, samples should be taken between days 5 and 9 of the menstrual cycle.[44] A serum human chorionic gonadotropin (hCG) level should be checked to rule out pregnancy in women with oligomenorrhea or amenorrhea.

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Screening Laboratory Studies

Late-onset congenital adrenal hyperplasia due to 21-hydroxylase deficiency can be ruled out by measuring serum 17-hydroxyprogesterone levels after a cosyntropin stimulation test. A 17-hydroxyprogesterone level of less than 1000 ng/dL—measured 60 minutes after cosyntropin stimulation—excludes late-onset congenital adrenal hyperplasia.

Women with PCOS should be screened for Cushing syndrome or acromegaly only if there is a clinical suspicion of these conditions. Cushing syndrome can be ruled out by checking a 24-hour urine sample for free cortisol and creatinine. Levels of urinary free cortisol that are 4 times the upper limit of normal are diagnostic for Cushing syndrome.[45] An overnight dexamethasone suppression test is also useful for screening for Cushing syndrome.

A serum insulin-like growth factor (IGF)–1 level should be checked to rule out acromegaly. Serum IGF-1 is a sensitive and specific marker of growth hormone (GH) excess. Normal levels rule out GH excess.

A small percentage of patients with PCOS have elevated prolactin levels (typically >25 mg/dL). Hyperprolactinemia can be excluded by checking a fasting serum prolactin concentration.

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Hormone Levels

Androgens

Androgen excess can be tested by measuring total and free testosterone levels or a free androgen index. An elevated free testosterone level is a sensitive indicator of androgen excess. Other androgens, such as dehydroepiandrosterone sulfate (DHEA-S), may be normal or slightly above the normal range in patients with PCOS. Levels of sex hormone–binding globulin (SHBG) are usually low in patients with PCOS.

Androstenedione levels are also elevated in women with PCOS. This androgen precursor is 60% ovarian and 40% adrenal in derivation.

Patients with androgen-secreting ovarian or adrenal tumors can present with hirsutism, amenorrhea, and signs of virilization. Although the clinical picture of symptom onset and progression is more predictive than androgen levels, their testosterone level may be greater than 150 ng/dL and their DHEA-S level may be above 800 mcg/dL. DHEA-S is derived from the adrenal gland, and therefore, elevation of DHEA-S would be suggestive of an adrenal origin.

Follicle-stimulating hormone and luteinizing hormone levels

The follicle-stimulating hormone (FSH) level should be checked to rule out primary ovarian failure. In patients with PCOS, FSH levels are within the reference range or low. Luteinizing hormone (LH) levels are elevated for Tanner stage, sex, and age. The LH-to-FSH ratio is usually greater than 3.

Stimulation testing with a long-acting gonadotropin-releasing hormone (GnRH) agonist induces a characteristic rise in ovarian-derived 17-hydroxyprogesterone after 24 hours. This is thought to be a result of excessive 17-hydroxylase activity.

Thyroid-stimulating hormone and free thyroxine levels

Thyroid dysfunction, rather than PCOS, may be the source of amenorrhea and hirsutism. In patients with PCOS, thyroid function tests are within the reference range.

Long-standing primary hypothyroidism can be associated with a markedly elevated circulating thyroid-stimulating hormone (TSH) level. Elevated alpha subunit delivery (from one half of the dimeric TSH molecule) can then cross-react with FSH and LH receptors on breast tissue, leading to premature thelarche and, on ovarian tissue, resulting in a PCOS–like picture. These physical findings of the van Wyk-Grumbach syndrome (ie, juvenile hypothyroidism, precocious puberty, and ovarian enlargement) resolve upon thyroxine replacement therapy.

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Glucose, Insulin, and Lipids

Because the prevalence of impaired glucose tolerance and type 2 diabetes mellitus is high in women with PCOS—particularly those who have a body mass index (BMI) greater than 30 kg/m2, have a strong family history of type 2 diabetes, or are older than 40 years—a 75-g oral glucose-tolerance test (OGTT) should be performed. A 2-hour postload glucose value of less than 140 mg/dL indicates normal glucose tolerance; a value of 140-199 mg/dL indicates impaired glucose tolerance; and a value of 200 mg/dL or higher indicates diabetes mellitus.[46]

Women diagnosed with prepregnancy PCOS should be screened for gestational diabetes before 20 weeks' gestation. These women have a higher rate of gestational diabetes than women in the general population; therefore, refer them for expert obstetric diabetic consultation if abnormal results are found.

Some women with PCOS have insulin resistance and an abnormal lipid profile (total cholesterol, >200 mg/dL; low-density lipoprotein [LDL] cholesterol, >160 mg/dL). Approximately one third of women with PCOS who are overweight have impaired glucose tolerance or type 2 diabetes mellitus by 30 years of age.[47]

A study concluded that insulin resistance and inflammatory markers may help identify adolescent girls with PCOS who are at the highest risk of developing the metabolic syndrome.[48] Metabolic heterogeneity also exists in women with PCOS according to phenotypic subgroup, with metabolic dysfunction confined to the subgroup with both oligomenorrhea and hyperandrogenic features.[49]

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Imaging for PCOS

Ultrasonography

Ovarian ultrasonography, preferably accomplished by using a transvaginal approach, can be performed to assess ovarian morphology. Polycystic ovarian morphology is defined in adults by the presence of 20 or more follicles per ovary and/or an ovarian volume of at least 10 mL in either ovary.[1]

Perform ultrasonography if the pelvic examination is inadequate, the patient has abdominal pain, testosterone levels are unusually high (eg, >200 ng/dL), it is needed to support the diagnostic criteria, or the patient is amenorrheic (to assess the endometrial thickness and exclude anatomic causes of amenorrhea). See the image below.

CT and MRI

If a tumor is suspected, obtain a computed tomography (CT) scan or magnetic resonance imaging (MRI) scan to visualize the adrenals and ovaries. MRI is an excellent method for imaging the ovaries and is a useful alternative in very obese women in whom the ovaries might not be visualized with transvaginal ultrasonography (TVUS) and in those patients in whom TVUS is inappropriate, such as adolescent girls.

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Histologic Findings

In patients with PCOS, histologic changes of the ovary include enlarged, sclerotic, multiple cystic follicles. See the image below.

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Treatment

Approach Considerations

Lifestyle changes, such as diet and exercise, are considered first-line treatment for adolescent girls and women with polycystic ovarian syndrome (PCOS).[50] Pharmacologic treatments are reserved for so-called metabolic derangements, such as anovulation, hirsutism, and menstrual irregularities.

Consultation with an endocrinologist is necessary for performing an adrenocorticotropic hormone (ACTH) stimulation test or for other causes of menstrual irregularity such as thyroid disease or pituitary adenoma. A reproductive endocrinologist should be consulted if the patient is infertile and desires pregnancy.[9]

The following are among the recommendations for management of PCOS from the 2023 International Evidence-based Guideline[1]:

- For management of irregular menstrual cycles and/or hirsutism, combined oral contraceptive pills are the first-line pharmacologic option
- No specific oral contraceptive is recommended; however, preparations with a lower dose of ethinyl estradiol and fewer adverse effects are preferred
- The use of metformin alone is recommended for adults who have a body mass index of 25 kg/m2 or higher primarily to manage the metabolic features of PCOS, such as insulin resistance and glucose and lipid levels
- Metformin is preferred over inositol, which has limited effects on hirsutism, weight, and ovulation in patients with PCOS

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Lifestyle Modifications

Lifestyle modifications such as weight loss and increased exercise in conjunction with a change in diet consistently reduce the risk of diabetes. This approach should be considered first-line treatment for patients with PCOS.[1,37,9] These modifications have been effective in restoring ovulatory cycles and achieving pregnancy in obese women with PCOS. Weight loss in obese women with PCOS also improves hyperandrogenic features.

Diet and activity

Patients with PCOS who have impaired glucose tolerance should start a comprehensive program of diet and exercise to reduce their risk of developing diabetes mellitus. Encourage moderate physical activity, provided the patient has no contraindications. Discourage smoking because of the increased risk of cardiovascular disease. In addition, obese women with PCOS can benefit from a low-calorie diet for weight reduction.

A diet patterned after the type 2 diabetes diet has been recommended for PCOS patients.[51] This diet emphasizes increased fiber; decreased refined carbohydrates, trans fats, and saturated fats; and increased omega-3 and omega-9 fatty acids. In some obese patients with PCOS, weight loss has improved menstrual regularity.[52]

A meta-analysis of 12 studies with 1529 participants found that weight loss interventions using medication or behavioral changes can reduce insulin resistance, decrease the free androgen index, and improve menstrual frequency in women with PCOS. Weight loss may not significantly reduce hirsutism or improve quality of life, however.[53]

A study by Jamilian et al found that soy isoflavone administration for 12 weeks in women with PCOS significantly improved markers of insulin resistance, hormonal status, triglycerides, and biomarkers of oxidative stress.[54]

Women with an abnormal lipid profile should be counseled on ways to manage the dyslipidemia. Such measures include eating a diet low in cholesterol and saturated fats and increasing physical activity.

Accumulating evidence suggests an association of vitamin D deficiency with PCOS.[55] One study found insufficient levels of 25-hydroxyvitamin D (< 30 ng/mL) in almost 75% of PCOS patients, with lower levels in those with metabolic syndrome (17.3 ng/mL) than in those without metabolic syndrome (25.8 ng/mL).[56]

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Drug Treatment

Medical management of PCOS is aimed at the treatment of metabolic derangements, anovulation, hirsutism, and menstrual irregularity. The use of insulin-sensitizing drugs to improve insulin sensitivity is associated with a reduction in circulating androgen levels, as well as improvement in both the ovulation rate and glucose tolerance.[37] ACOG notes that eflornithine in conjunction with laser treatment is superior to laser therapy alone in treating hirsutism.[37]

Oral contraceptives

First-line medical therapy usually consists of an oral contraceptive to induce regular menses. Oral contraceptives containing a combination of estrogen and progestin increase sex hormone–binding globulin (SHBG) levels and thereby reduce the free testosterone level. Luteinizing hormone (LH) and follicle-stimulating hormone (FSH) levels are also suppressed. This restores cyclic exposure of the endometrium to estrogen-progestin, with the resumption of menstrual periods and decreased hirsutism. However, the use of oral contraceptives may be associated with an increased risk of thrombosis and metabolic abnormalities.

ACOG recommends use of combination low-dose hormonal contraceptive agents for long-term management of menstrual dysfunction.[37] If symptoms such as hirsutism are not sufficiently alleviated, an androgen-blocking agent may be added.

Withdrawal bleeding can be induced with medroxyprogesterone given for 5-10 days before the start of oral contraceptive therapy. Pregnancy should be excluded before therapy with oral contraceptives or androgen-blocking agents is started.

Letrozole and clomiphene citrate

Letrozole is the recommended first-line pharmacologic agent for ovulation induction in women with PCOS and infertility, as it results in higher live-birth and ovulation rates compared with clomiphene citrate. Clomiphene remains an acceptable alternative if letrozole is unavailable or contraindicated.[1,37]

A double-blind trial by Legro et al found that letrozole is more effective than clomiphene in the treatment of infertility in PCOS. Based on treatment periods of up to five cycles, the study, which involved 750 anovulatory women with PCOS, found that the birth rates for letrozole and clomiphene were 27.5% and 19.1%, respectively. The rate of congenital abnormalities and the risk of pregnancy loss in the letrozole and clomiphene groups were found to be comparable, although the likelihood of twin births was lower with letrozole.[57]

Metformin

If the patient develops type 2 diabetes mellitus, consider treatment with oral antihyperglycemic drugs, such as metformin. Metformin can also be considered in other women with PCOS who are insulin resistant and therefore at risk of developing cardiovascular disease, even women without type 2 diabetes mellitus.

Clinical trials have shown that metformin can effectively reduce androgen levels, improve insulin sensitivity, and facilitate weight loss in patients with PCOS as early as adolescence.[58] One study concluded that the use of metformin throughout pregnancy was associated with a 9-fold decrease in gestational diabetes in women with PCOS.[59] In addition to having the potential to reduce gestational diabetes in pregnant women with PCOS, metformin may also reduce the risk of preeclampsia in this population.[60]

A long-term study suggested that metformin continued to improve the metabolic profile of women with PCOS over a 36-month treatment course, particularly improving circulating high-density lipoprotein cholesterol (HDL-C), diastolic blood pressure, and body mass index (BMI).[61] However, data are insufficient as yet to recommend metformin to all women with PCOS.

Other agents

If the patient has concomitant adrenal hyperandrogenism, treatment with low-dose prednisone or dexamethasone may be considered.

Depot leuprolide acetate is effective in suppressing ovarian hormone production, which effectively induces menopause; therefore, this drug must be accompanied by hormone therapy. This treatment approach has not gained widespread favor.

Several medications, including benzoyl peroxide, topical retinoids, and topical and oral antibiotics, are effective for acne treatment. Systemic isotretinoin is used for severe or refractory cases.

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Metabolic Derangements

In patients with PCOS who are obese, endocrine-metabolic parameters markedly improve after 4-12 weeks of dietary restriction. Their sex hormone–binding globulin (SHBG) levels rise, and free testosterone levels fall by 2-fold.[62] Serum insulin and insulin-like growth factor-1 (IGF-1) levels also decrease. In patients with PCOS who are obese, weight loss is associated with a reduction of hirsutism and a return of ovulatory cycles in 30% of women, thereby improving pregnancy rates, as well as improving glucose tolerance and lipid levels.[10,37]

The Androgen Excess and Polycystic Ovary Syndrome Society recommends lifestyle management as the primary therapy for metabolic complications in overweight and obese women with PCOS.[63] A moderate amount of daily exercise increases levels of IGF-1 binding protein and decreases levels of IGF-1 by 20%. Modest weight loss of 2-5% of total body weight can help restore ovulatory menstrual periods in obese patients with PCOS. A decrease of 500-1000 calories daily, along with 150 minutes of exercise per week, can cause ovulation.

Metformin, an antidiabetic drug, improves insulin resistance and decreases hyperinsulinemia in patients with PCOS.[64] This drug also has a small but beneficial effect on metabolic syndrome, as well as potentially causing a modest reduction in androgen levels (11%).[65] Note that women with a body mass index (BMI) greater than 37 kg/m2 may not have a good response to metformin.[65] An Italian study of 33 patients with PCOS demonstrated that metformin affected thyroid hormone levels by lowering thyroid-stimulating hormone (TSH) in hypothyroid patients with PCOS, regardless of whether these individuals received levothyroxine or were untreated.[66]

Ascertain that kidney and liver function are normal and that the patient does not have advanced congestive heart failure before starting metformin therapy. The usual starting dose is 500 mg given orally twice a day. Because common adverse effects are nausea, vomiting, and diarrhea, metformin should be taken with meals. Patients who develop these adverse effects can be instructed to decrease the dosage to once a day for a week and then gradually increase the dosage. Also, inform patients that there is a high likelihood that they will have ovulatory cycles while taking metformin. The US Food and Drug Administration (FDA) has not approved metformin for this indication.

A meta-analysis that included 12 studies involving women with PCOS showed that those who continued metformin throughout the first trimester experienced a lower risk for miscarriage (odds ratio [OR], 0.44) and higher clinical pregnancy rates (OR, 1.16) and live birth rates (OR, 1.14) than those who stopped metformin at the positive pregnancy test. Moreover, women who stopped metformin at pregnancy confirmation had a higher risk for miscarriage (OR, 1.46) than those who received either placebo or no treatment.[67]

However, a secondary analysis of two randomized, double-blind, placebo-controlled trials that included 182 children of mothers with PCOS reported that children exposed to metformin had higher BMI and increased prevalence of overweight/obesity at 4 years of age, compared with children in the placebo group. [68]

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Anovulation

Letrozole is recommended as first-line therapy to stimulate ovulation when fertility is desired.[1] An alternative pharmacologic therapy to stimulate ovulation is clomiphene citrate.[57]

Second-line therapy, when letrozole or clomiphene fails to lead to pregnancy, is either exogenous gonadotropins or laparoscopic ovarian surgery.[37] If gonadotropins are used, a low-dose regimen is recommended, and patients must be monitored with ultrasonography and laboratory studies. Note that gonadotropin therapy is expensive and is associated with an increased risk of multiple pregnancy and ovarian hyperstimulation syndrome.

Evidence suggests that metformin frequently, but not universally, improves ovulation rates and pregnancy rates in women with PCOS, especially in obese women.[1,37] In addition, pretreatment with metformin has been shown to enhance the efficacy of clomiphene for inducing ovulation.[69] Consider the combination of metformin and clomiphene in older women with visceral obesity and clomiphene resistance.[70] However, this combination does not significantly improve the live birth rate relative to clomiphene monotherapy.[70]

Patients with PCOS who are infertile but desire pregnancy should be referred to a reproductive endocrinologist for further evaluation and management of infertility. Morbidly obese women with PCOS should also be referred for pregnancy risk; metabolic surgery may be considered in morbidly obese women with PCOS, because many features of this syndrome are

reversible with successful weight loss. In vitro fertilization (IVF) is reserved for women with PCOS and unsuccessful gonadotropin therapy or those with other indications for this procedure.

A study by Chen et al found that among infertile women with PCOS, frozen-embryo transfer was associated with a higher rate of live birth, a lower risk of the ovarian hyperstimulation syndrome, and a higher risk of preeclampsia after the first transfer than was fresh-embryo transfer.[71]

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Hirsutism

A clear primary treatment for hirsutism in women with PCOS remains lacking.[37] However, short-term, nonpharmacologic treatments of hirsutism include shaving and the use of chemical depilatories and/or bleaching cream.[72] Plucking or waxing unwanted hair can result in folliculitis and ingrown hairs. Long-term, more permanent measures for unwanted hairs include electrolysis and laser treatment.

Adjunctive eflornithine with laser treatment is superior to laser therapy alone in treating hirsutism.[73] Eflornithine is a topical cream that can be used to slow hair growth. This agent works by inhibiting ornithine decarboxylase, which is essential for the rapidly dividing cells of hair follicles.

Weight reduction decreases androgen production in women who are obese; therefore, losing weight might slow hair growth.

Women who do not wish to become pregnant can be effectively treated for hirsutism with oral contraceptives.[74] Oral contraceptives slow hair growth in 60-100% of women with hyperandrogenemia.

Therapy can be started with a preparation that has a low dose of estrogen and a nonandrogenic progestin. Preparations that have norgestrel and levonorgestrel should be avoided because of their androgenic activity. There is also a risk of thrombotic events in obese women who use oral contraceptives; therefore, the proper precautions should be exercised to prevent such events. Oral contraceptives containing cyproterone acetate are also very effective in the treatment of more severe hirsutism[75]; however, this combination of agents has not been approved by the FDA for use in the United States.

Antiandrogens, such as spironolactone, are effective for hirsutism.[76] Spironolactone (50-100 mg twice daily) is an effective primary therapy for hirsutism. Because of the potential teratogenic effects of spironolactone, patients require an effective form of contraception (eg, an oral contraceptive). Adverse effects of spironolactone include gastrointestinal discomfort and irregular menstrual bleeding, which can be managed by adding an oral contraceptive.

Ovulation induction with clomiphene citrate, metformin, or both does not alter hirsutism in infertile hirsute women with PCOS.[77]

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Surgical Intervention

Surgical management of PCOS is aimed mainly at restoring ovulation. Ovarian wedge resection has fallen out of favor because of postoperative adhesion formation and the successful introduction of ovulation-inducing medications.

Various laparoscopic methods, including electrocautery, laser drilling, and multiple biopsy, have been used with the goal of creating focal areas of damage in the ovarian cortex and stroma. Laparoscopic ovarian drilling may be considered in women with clomiphene-resistant PCOS, especially in the presence of other laparoscopic indications.[70] A small French study also suggested that surgical management via ovarian drilling with hydrolaparoscopy may be beneficial in cases of PCOS that are resistant to clomiphene citrate.[78]

Potential complications must be considered as well. These include formation of adhesions and ovarian atrophy. Multiple pregnancy rates are lower with ovarian drilling than with gonadotropin treatment (1% vs 16%, respectively), but there are ongoing concerns about the long-term effects of ovarian drilling on ovarian function.[79]

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Long-Term Monitoring

PCOS is a disease with many long-term complications. Patients need regular follow-up with their physicians for early detection and management of any untoward sequelae associated with the syndrome.

Women with PCOS who conceive are at increased risk for gestational diabetes, preeclampsia, cesarean delivery, and preterm and postterm delivery. In addition, their newborns are at increased risk of being large for gestational age, but they are not at increased risk of stillbirth or neonatal death.[80]

Participation in a peer support group may alleviate distress and improve self-management.[81]

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Guidelines

Guidelines Summary

The following organizations have released guidelines for the management of polycystic ovarian syndrome (PCOS). Key diagnostic and treatment recommendations have been reviewed and integrated throughout the article.

- 2023. International PCOS Network. Recommendations from the 2023 International Evidence-based Guideline for the Assessment and Management of Polycystic Ovary Syndrome[1]
- 2018. American College of Obstetricians and Gynecologists' Committee on Practice Bulletins—Gynecology. ACOG Practice Bulletin No. 194: Polycystic Ovary Syndrome[37]
- 2018. Society of Obstetricians and Gynaecologists of Canada. No. 362-Ovulation Induction in Polycystic Ovary Syndrome[9]

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Medication

Antiandrogens

Finasteride (Propecia, Proscar)

Leuprolide (Camcevi, Camcevi ETM, Eligard)

Spironolactone (Aldactone, CaroSpir)

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Aromatase Inhibitors

Letrozole (Femara)

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Hypoglycemic Agents

Insulin isophane human/insulin regular human (Humulin 70/30, Novolin 70/30)

Metformin (Glumetza)

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Oral Contraceptives

Drospirenone/ethinyl estradiol/levomefolate (Ethinyl estradiol, Beyaz, Safyral)

Medroxyprogesterone (Depo-SubQ Provera 104, DepoProvera, MPA)

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Selective Estrogen Receptor Modulators

Clomiphene (Clomid, Serophene)

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Topical Hair-Removal Agents

Eflornithine cream

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Questions & Answers

Overview

What is polycystic ovarian syndrome (PCOS)?

What are the signs and symptoms of polycystic ovarian syndrome (PCOS)?

Which physical findings suggest polycystic ovarian syndrome (PCOS)?

Which lab tests are performed in the workup of polycystic ovarian syndrome (PCOS)?

Which imaging studies are performed in the workup of polycystic ovarian syndrome (PCOS)?

Which procedures are performed in the workup of polycystic ovarian syndrome (PCOS)?

What is the first-line treatment for polycystic ovarian syndrome (PCOS)?

What are the pharmacologic treatments for polycystic ovarian syndrome (PCOS)?

What are the surgical options for treating polycystic ovarian syndrome (PCOS)?

What are the signs and symptoms of polycystic ovarian syndrome (PCOS)?

What are the diagnostic criteria for polycystic ovarian syndrome (PCOS)?

What is the role of androgens and estrogen metabolism in the etiology of polycystic ovarian syndrome (PCOS)?

What is the role of insulin resistance in the etiology of polycystic ovarian syndrome (PCOS)?

What is the role of anovulation in the etiology of polycystic ovarian syndrome (PCOS)?

What is the role of hyperinsulinemia in the etiology of polycystic ovarian syndrome (PCOS)?

How are ovaries characterized in polycystic ovarian syndrome (PCOS)?

What functional abnormality may be found in women with polycystic ovarian syndrome (PCOS)?

What is the role of genetics in the etiology of polycystic ovarian syndrome (PCOS)?

Which gene linked to obesity may be an etiologic factor in polycystic ovarian syndrome (PCOS)?

Which genes have been associated with polycystic ovarian syndrome (PCOS)?

What is the prevalence of polycystic ovarian syndrome (PCOS) in the US?

What is the ethnic variability for hirsutism among women with polycystic ovarian syndrome (PCOS)?

In what age group is polycystic ovarian syndrome (PCOS) commonly diagnosed?

Polycystic ovarian syndrome (PCOS) increases the risk of developing which conditions?

What is the prevalence of insulin resistance in women with polycystic ovarian syndrome (PCOS)?

What are the AACE and the ACE recommendations for diabetes screening in women with polycystic ovarian syndrome (PCOS)?

What is the risk for endometrial hyperplasia and carcinoma in patients with polycystic ovarian syndrome (PCOS)?

What is the risk for breast or ovarian cancer in patients with polycystic ovarian syndrome (PCOS)?

What education should be given to patients with polycystic ovarian syndrome (PCOS)?

Presentation

Which family history suggests polycystic ovarian syndrome (PCOS)?

Which menstrual irregularities suggest polycystic ovarian syndrome (PCOS)?

Which manifestations of hyperandrogenism suggest polycystic ovarian syndrome (PCOS)?

What are the ACOG recommendations for screening with 17-hydroxyprogesterone levels in women with polycystic ovarian syndrome (PCOS)?

What effect does polycystic ovarian syndrome (PCOS) have on fertility?

What is the prevalence of obesity among women with polycystic ovarian syndrome (PCOS)?

What is the prevalence of metabolic syndrome among women with polycystic ovarian syndrome (PCOS)?

Which cardiovascular disorders are at increased risk in women with polycystic ovarian syndrome (PCOS)?

What are the ACOG recommendations for diabetes screening in women with polycystic ovarian syndrome (PCOS)?

How prevalent is sleep apnea among women with polycystic ovarian syndrome (PCOS)?

Which findings of hyperandrogenism suggest polycystic ovarian syndrome (PCOS)?

What is the modified Ferriman-Gallwey (mFG) scoring system and how is it used in the evaluation of suspected polycystic ovarian syndrome (PCOS)?

How prevalent is abdominal obesity in women with polycystic ovarian syndrome (PCOS)?

What causes acanthosis nigricans in women with polycystic ovarian syndrome (PCOS)?

How is acanthosis nigricans staged in women with polycystic ovarian syndrome (PCOS)?

What does a finding of high blood pressure suggest in women with polycystic ovarian syndrome (PCOS)?

What are the physical findings of ovaries in polycystic ovarian syndrome (PCOS)?

DDX

Which disorders should be included in the differential diagnosis of polycystic ovarian syndrome (PCOS)?

What are the differential diagnoses for Polycystic Ovarian Syndrome?

Workup

How is polycystic ovarian syndrome (PCOS) diagnosed?

What baseline screening tests are recommended by the RCOG for women with suspected polycystic ovarian syndrome (PCOS)?

When is an infertility evaluation indicated in women with suspected polycystic ovarian syndrome (PCOS)?

When should lab samples be obtained for the workup of polycystic ovarian syndrome (PCOS)?

How is late-onset congenital hyperplasia due to 21-hydroxilase deficiency differentiated from polycystic ovarian syndrome (PCOS)?

When is screening for Cushing syndrome or acromegaly indicated in the workup of polycystic ovarian syndrome (PCOS)?

How is hyperprolactinemia differentiated from polycystic ovarian syndrome (PCOS)?

What is the role of androgen level testing in the workup of polycystic ovarian syndrome (PCOS)?

How are androgen-secreting ovarian or adrenal tumors differentiated from polycystic ovarian syndrome (PCOS)?

What is the role of follicle-stimulating hormone (FSH) level testing in the workup of polycystic ovarian syndrome (PCOS)?

How is hypothyroidism differentiated from polycystic ovarian syndrome (PCOS)?

What is the role of oral glucose-tolerance testing in the workup of polycystic ovarian syndrome (PCOS)?

When should pregnant women with polycystic ovarian syndrome (PCOS) be screened for gestational diabetes?

What does insulin resistance and inflammatory markers suggest in adolescents with polycystic ovarian syndrome (PCOS)?

What is the role of ultrasonography in the workup of polycystic ovarian syndrome (PCOS)?

When are CT scanning and MRI indicated in the workup of polycystic ovarian syndrome (PCOS)?

Which histologic changes in ovaries suggest polycystic ovarian syndrome (PCOS)?

Treatment

What is the first-line treatment for polycystic ovarian syndrome (PCOS)?

What do elevated mean platelet volume (MPV) levels suggest in women with polycystic ovarian syndrome (PCOS)?

Which specialists should be consulted in the management of polycystic ovarian syndrome (PCOS)?

What are the Endocrine Society practice guidelines for the diagnosis and treatment of polycystic ovarian syndrome (PCOS)?

How effective are lifestyle changes in the treatment of polycystic ovarian syndrome (PCOS)?

What are the goals of medical management of polycystic ovarian syndrome (PCOS)?

What is the first line medical therapy for polycystic ovarian syndrome (PCOS)?

How is infertility treated in women with polycystic ovarian syndrome (PCOS)?

What is the role of metformin in the treatment of polycystic ovarian syndrome (PCOS)?

What is the role of corticosteroids in the treatment of polycystic ovarian syndrome (PCOS)?

What is the role of leuprolide acetate (Lupron) in the treatment of polycystic ovarian syndrome (PCOS)?

Which medications can be used to treat acne in women with polycystic ovarian syndrome (PCOS)?

What are the AE-PCOS recommendations for the treatment of metabolic complications of polycystic ovarian syndrome (PCOS)?

What is the role of metformin in the treatment of polycystic ovarian syndrome (PCOS)?

What do the ACOG and SOGC recommend as first-line therapy for ovulation stimulation in women with polycystic ovarian syndrome (PCOS)?

What is the second-line therapy for ovulation stimulation in women with polycystic ovarian syndrome (PCOS)?

Which drugs are effective in treating anovulation in women with polycystic ovarian syndrome (PCOS)?

What are the treatment options for infertility in women with polycystic ovarian syndrome (PCOS)?

What is the role of frozen-embryo transfer for the treatment of infertility in polycystic ovarian syndrome (PCOS)?

What are the treatment options for hirsutism secondary to polycystic ovarian syndrome (PCOS)?

What is the role of contraceptives in the treatment of hirsutism secondary to polycystic ovarian syndrome (PCOS)?

What is the role of antiandrogens for the treatment of hirsutism secondary to polycystic ovarian syndrome (PCOS)?

Is clomiphene citrate or metformin effective in reducing hirsutism secondary to polycystic ovarian syndrome (PCOS)?

What is the role of surgical intervention for the treatment of polycystic ovarian syndrome (PCOS)?

What monitoring is needed for women with polycystic ovarian syndrome (PCOS)?

What conditions are women with polycystic ovarian syndrome (PCOS) at increased risk for during pregnancy?

What are the benefits of peer support groups in the treatment of polycystic ovarian syndrome (PCOS)?

What are the FDA warnings for the use of statins?

How can women with polycystic ovarian syndrome (PCOS) lower their risk for developing diabetes mellitus?

What is the role of soy isoflavone in the treatment of women with polycystic ovarian syndrome (PCOS)?

How is dyslipidemia treated in women with polycystic ovarian syndrome (PCOS)?

Does vitamin D deficiency increase the risk for metabolic syndrome in women with polycystic ovarian syndrome (PCOS)?

Guidelines

What are the AACE, ACE, and AE-PCOS guidelines for treatment and management of polycystic ovarian syndrome (PCOS)?

Medications

Which medications are used to treat polycystic ovarian syndrome (PCOS)?

What FDA warnings exist for medications used in the treatment of polycystic ovarian syndrome (PCOS)?

Which medications in the drug class Topical Hair-Removal Agents are used in the treatment of Polycystic Ovarian Syndrome?

Which medications in the drug class Selective Estrogen Receptor Modulators are used in the treatment of Polycystic Ovarian Syndrome?

Which medications in the drug class Oral Contraceptives are used in the treatment of Polycystic Ovarian Syndrome?

Which medications in the drug class Hypoglycemic Agents are used in the treatment of Polycystic Ovarian Syndrome?

Which medications in the drug class Aromatase Inhibitors are used in the treatment of Polycystic Ovarian Syndrome?

Which medications in the drug class Antiandrogens are used in the treatment of Polycystic Ovarian Syndrome?

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